Is it possible to predict how often people from different countries will participate in cultural activities?

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1. Introduction

Eurostat (European Statistical Office) is a Directorate-General of the European Commission whose main responsibilities are to provide statistical information to the institutions of the European Union and to promote the harmonisation of statistical methods across its member states.

My goal is to predict the frequency of segmented population groups' participation in cultural activities (cinema, live performances or cultural sites) based on their personal traits and socioeconomic status.

Because participation frequency is aggregated in Eurostat's datasets into groups (*Not in the last 12 months*, *From 1 to 3 times* [last year], *At least 4 times* [last year]), this is a classification goal.

2. Data

For this project I used 2 different datasets. The first dataset contains the aggregated information of the population grouped by education level, age, sex and geopolitical entity as well as the most "popular" participation group (N = 1504, see **Figure 1**). For example, most of the women aged from 16 to 24 from Albania with ISCED level between 0 and 2 participated in cultural activities from 1 to 3 times in a year. The second dataset contains the information of the population grouped by degree of urbanisation, household composition, income quintile and geopolitical entity as well as the most "popular" participation group (N = 3618, see **Figure 2**).

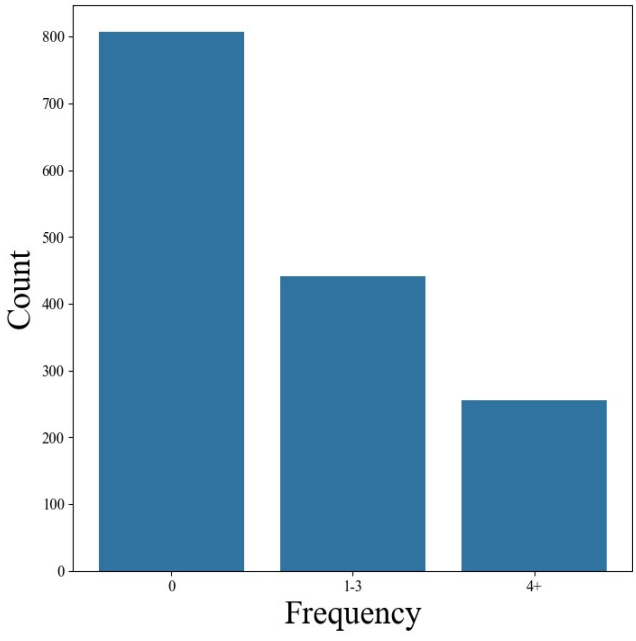


Figure 1 – distribution of dependent variable

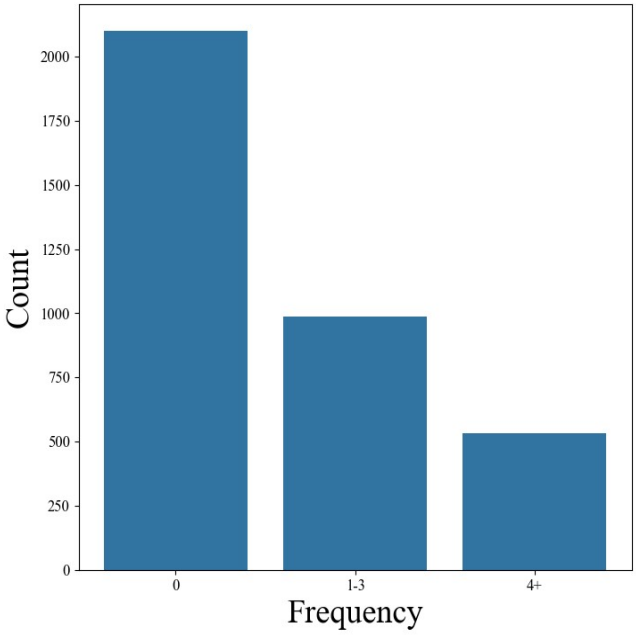


Figure 2 – distribution of dependent variable

3. Data manipulation

I cleaned both datasets from technical columns, removed overlapping aggregated groups and transformed categorial variables into numerical. Then, I grouped countries into five regions (East, North, South, West, Balkans) and transformed them with one-hot encoding. Finally, for each socioeconomic group I left only the frequency group with the highest percentage.

4. Machine learning

I separated both datasets into training set (75%) and testing set (25%). I then built and evaluated three classification models:
-Logistic regression
-Random forest
-Gradient Boosting

For random forest and gradient boosting models I chose best parameters by using GridSearchCV algorithm.

5. Results

5.1. Predicting frequency group based on age, sex, education level and geopolitical entity

5.1.1. Logistic regression

The model achieves 75.1% accuracy on the training set and 76.3% on the testing set (see Table 1). The negligible gap between the two scores suggests low variance. However, the moderate overall accuracy indicates that the model may suffer from bias.

	Predicted 0	**Predicted 1-3**	**Predicted 4+**
Actual 0	177	17	1
Actual 1-3	20	80	22
Actual 4+	4	25	30

Table 1 - Confusion matrix of the test dataset for the logistic regression model applied to df_educational. The accuracy is 76.3%.

5.1.2. Random Forest

The model achieves 80.6% accuracy on the training set and 81.6% on the testing set (see Table 2). The small positive gap — where test accuracy slightly exceeds training accuracy — indicates low variance and good generalization.

	Predicted 0	**Predicted 1-3**	**Predicted 4+**
Actual 0	182	12	1
Actual 1-3	16	92	14
Actual 4+	3	23	33

Table 2 - Confusion matrix of the test dataset for the Random Forest model applied to df_educational. The accuracy is 81.6%.

5.1.3. Gradient Boosting

The model achieves 80.2% accuracy on the training set and 81.6% on the testing set (see Table 3). Similarly to the Random Forest model, the slight difference between training and test accuracy confirms low variance and good generalization.

	Predicted 0	**Predicted 1-3**	**Predicted 4+**
Actual 0	180	13	2
Actual 1-3	14	94	14
Actual 4+	1	25	33

Table 3 - Confusion matrix of the test dataset for the Gradient Boosting model applied to df_educational. The accuracy is 81.6%.

5.2. Predicting frequency group based on degree of urbanisation, household composition, income quintile and geopolitical entity

5.2.1. Logistic regression

The model achieves 77.9% accuracy on the training set and 77.8% on the testing set (see Table 4). The near-identical scores indicate very low variance. However, the moderate overall accuracy suggests that the model may suffer from bias.

	Predicted 0	**Predicted 1-3**	**Predicted 4+**
Actual 0	491	31	16
Actual 1-3	65	142	29
Actual 4+	25	35	71

Table 4 - Confusion matrix of the test dataset for the logistic regression model applied to df_economical. The accuracy is 77.8%.

5.2.2. Random Forest

The model achieves 79.7% accuracy on the training set, but its accuracy drops to 76.5% on the testing set (see Table 5). The moderate gap between the two scores suggests some degree of variance.

	Predicted 0	**Predicted 1-3**	**Predicted 4+**
Actual 0	480	50	8
Actual 1-3	62	149	25
Actual 4+	26	42	63

Table 5 - Confusion matrix of the test dataset for the Random Forest model applied to df_economical. The accuracy is 76.5%.

5.2.3. Gradient Boosting

The model achieves 79.0% accuracy on the training set and 77.5% on the testing set (see Table 6). The moderate gap between the two scores suggests some degree of variance.

	Predicted 0	**Predicted 1-3**	**Predicted 4+**
Actual 0	486	41	11
Actual 1-3	59	149	28
Actual 4+	24	41	66

Table 6 - Confusion matrix of the test dataset for the Gradient Boosting model applied to df_economical. The accuracy is 77.5%.

6. Results

For predicting predicting frequency group based on age, sex, education level and geopolitical entity, the best model is the Random forest with an accuracy of 81.6%. However, when it comes to predicting frequency group based on degree of urbanisation, household composition, income quintile and geopolitical entity, the best model is the Logistic regression with an accuracy of 77.8%.